Main Information for Low-Altitude Targets

In this work, we construct the calibration point clouds for 10 types of UAVs, 10 types of birds, 10 types of vehicles, and 10 types of pedestrians. The UAVs include DJI Mini 2, DJI Mini 4 Pro, DJI Avata, DJI Mavic 3 Pro, DJI Mavic 4 Pro, DJI Inspire 2, TypeA Firefighting UAV, Type-B Firefighting UAV, DJI Matrice 600 Pro, and EH 216 S. The birds include crow, eagle, kingfisher, parrot, pigeon, ramphastidae, sparrow, seagull, owl, and wild goose. The vehicles include AITO M9 2024, Tesla Roadster 2020, Xiaomi SU7 2024, XPeng G9 2022, ZEEKR 001 2022, Audi A6L 2019, BYD Song L EV 2024, Geely Galaxy L7 2024, Leading Ideal L8 2023, and NIO ET5T 2023. The pedestrians include 8 walking pedestrians and 2 running pedestrians. The calibration point clouds and detailed information of these low-altitude targets are explained as follows.

Readers can see the motion point clouds of these 40 types of low-altitude targets in https://github.com/Alivn999/COSMOS-Networked-ISAC-Enabled-Target-Recognition-Towards-Low-Altitude-Economy/tree/main/Dynamic_Videos_for_40_Types_of_Low_Altitude_Targets. It is worth noting that the displayed dynamic video is only the motion point cloud under a set of motion parameters. When constructing the massive echo signal dataset, the motion parameters of each sample are random, and hence the motion point cloud of each sample is also different. This ensures the richness of the echo signal dataset for low-altitude target recognition.

The main information of these 40 types of low-altitude targets are as follows.

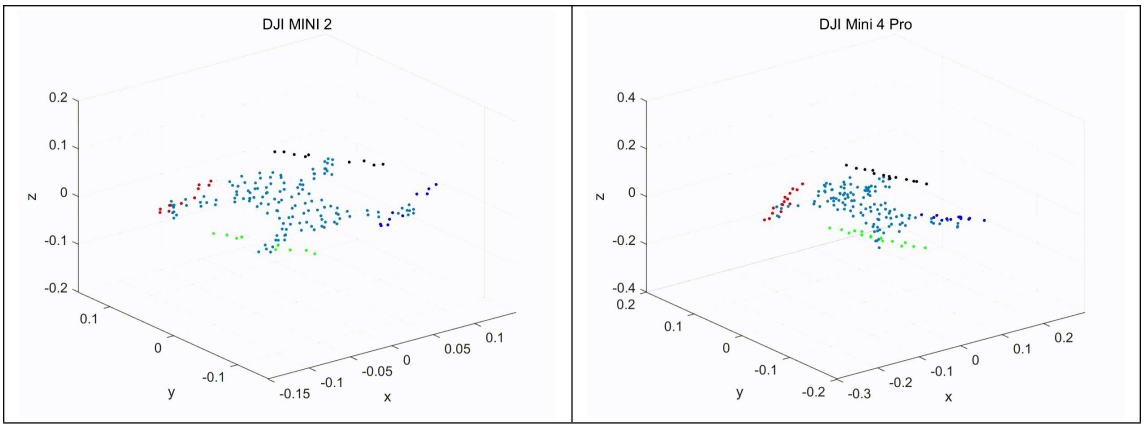
Main information of UAV targets.

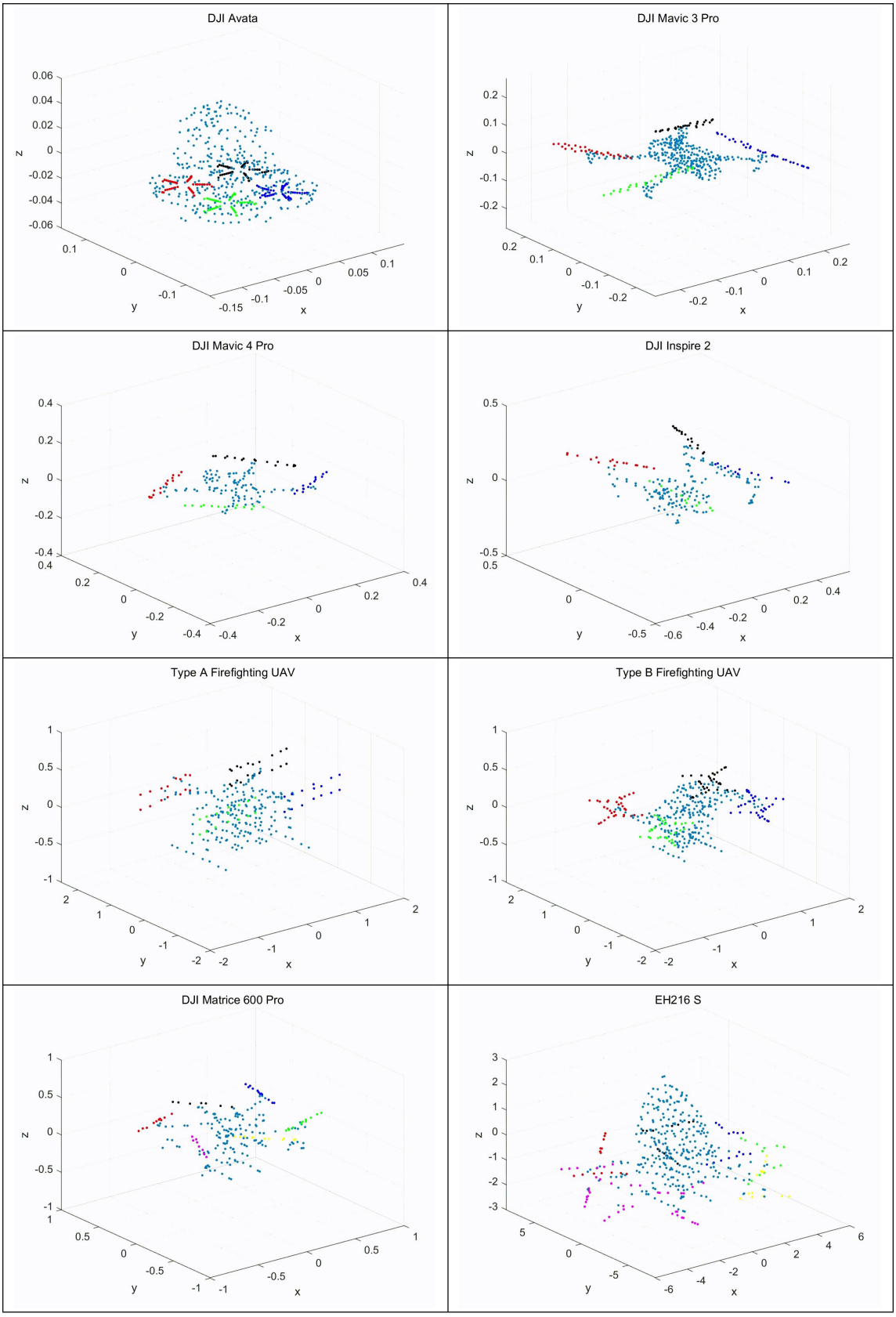
Table 1: Main information of the UAV targets.

Name	Category	Size (m)	Paddles Number	Paddles Frequency (Hz)	3D Velocity Range (m/s)
DJI Mini 2	UAV	0.2192	4	80~256	(±10, ±10, ±3)
DJI Mini 4 Pro	UAV	0.2397	4	60~100	(±13, ±13, ±6)
DJI Avata	UAV	0.2048	4	90~660	(±20, ±20, ±9)
DJI Mavic 3 Pro	UAV	0.3815	4	60~100	(±15, ±15, ±7)
DJI Mavic 4 Pro	UAV	0.4198	4	60~120	(±18, ±18, ±10)
DJI Inspire 2	UAV	0.6844	4	30~120	(±20, ±20, ±5)
Type-A Firefighting UAV	UAV	2.3118	4	60~180	(±12, ±12, ±6)
Type-B Firefighting UAV	UAV	2.3834	4	80~180	(±13, ±13, ±10)
DJI Matrice 600 Pro	UAV	1.1365	6	50~120	(±10, ±10, ±5)
EH 216 S	UAV	5.2704	8	30~60	(±25, ±25, ±2)

Table 1 presents the main information of UAV targets. The size of a UAV target refers to the approximate length of the diagonal of its body. The paddle frequency refers to the number of rotations of the UAV's paddle within 1 s. The 3D velocity range refers to the value of the 3D velocity not exceeding (v_x, v_y, v_z).

The calibration point clouds of the UAV targets are as follows.




DJI Avata
DJI Mavic 3 Pro
DJI Mavic 4 Pro
DJI Inspire 2
Type A Firefighting UAV
Type B Firefighting UAV
DJI Matrice 600 Pro
EH216 S
x
y
z

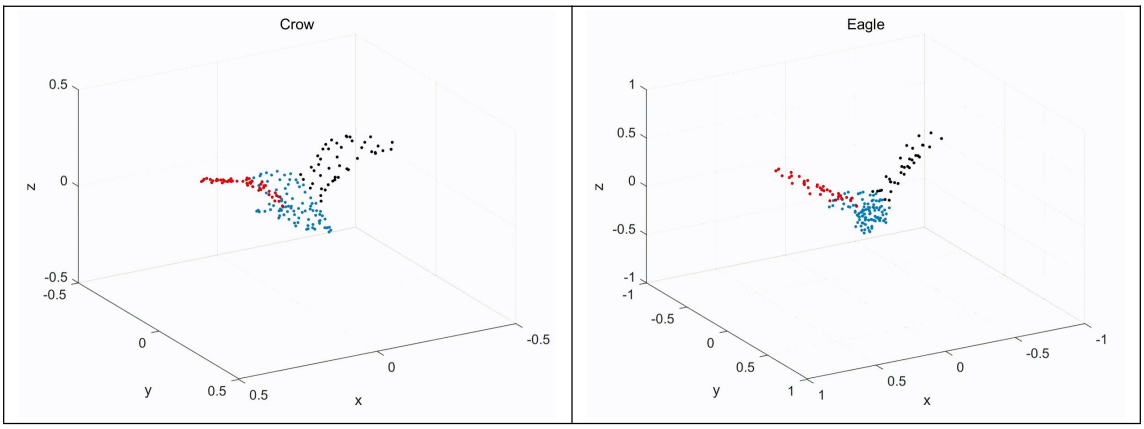
Main information of bird targets.

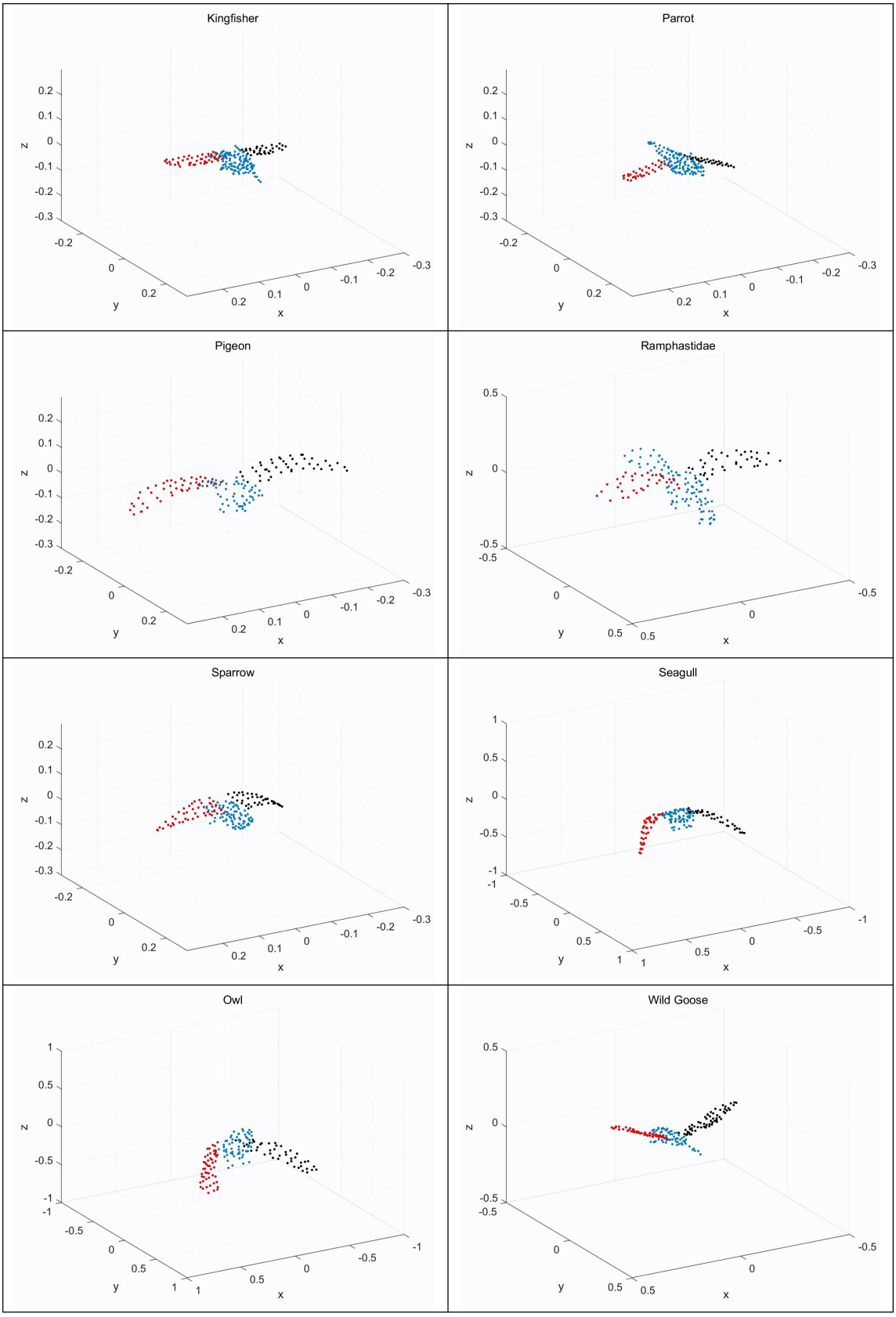
Table 2: Main information of the bird targets.

Name	Category	Size (m)	Wings Number	Wings Frequency (Hz)	3D Velocity Range (m/s)
Crow	bird	0.3480	2	3~5	(±8, ±8, ±7)
Eagle	bird	0.4325	2	2~4	(±9, ±9, ±8)
Kingfisher	bird	0.1519	2	6.5~8.5	(±7, ±7, ±6)
Parrot	bird	0.2383	2	9~11	(±7, ±7, ±6)
Pigeon	bird	0.2196	2	4.5~6	(±8, ±8, ±7)
Ramphastidae	bird	0.5032	2	6~8	(±7, ±7, ±6)
Sparrow	bird	0.1385	2	17~19	(±7, ±7, ±6)
Seagull	bird	0.3782	2	2.5~3.5	(±8, ±8, ±7)
Owl	bird	0.3843	2	3~5	(±7, ±7, ±6)
Wild goose	bird	0.2994	2	2~4	(±8, ±8, ±7)

Table 2 presents the main information of bird targets. The size of a bird target refers to the approximate length of its body from head to toe. The wing frequency refers to the number of times a bird flaps its wings within 1 s. The 3D velocity range refers to the value of the 3D velocity not exceeding (v_x, v_y, v_z).

The calibration point clouds of the bird targets are as follows.




Kingfisher
Parrot
Pigeon
Ramphastidae
Sparrow
Seagull
Owl
Wild Goose
x
y
z

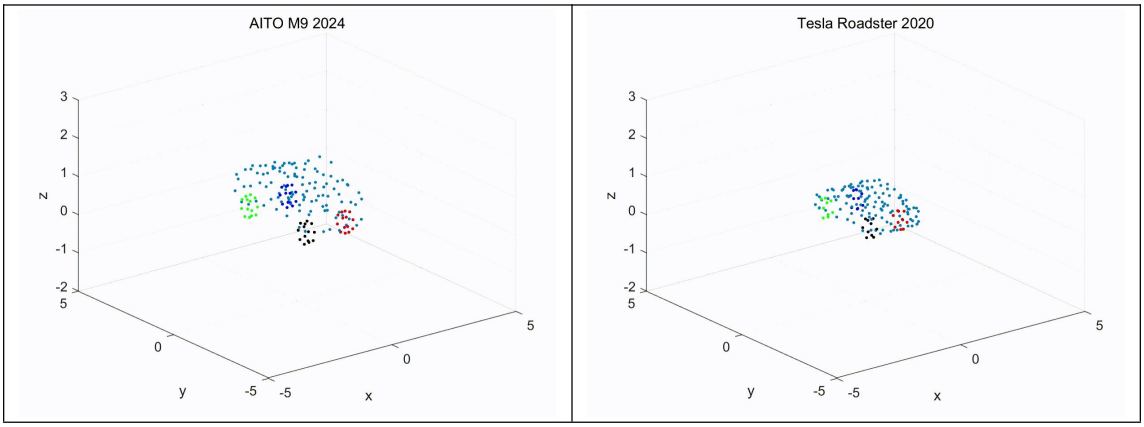
Main information of vehicle targets.

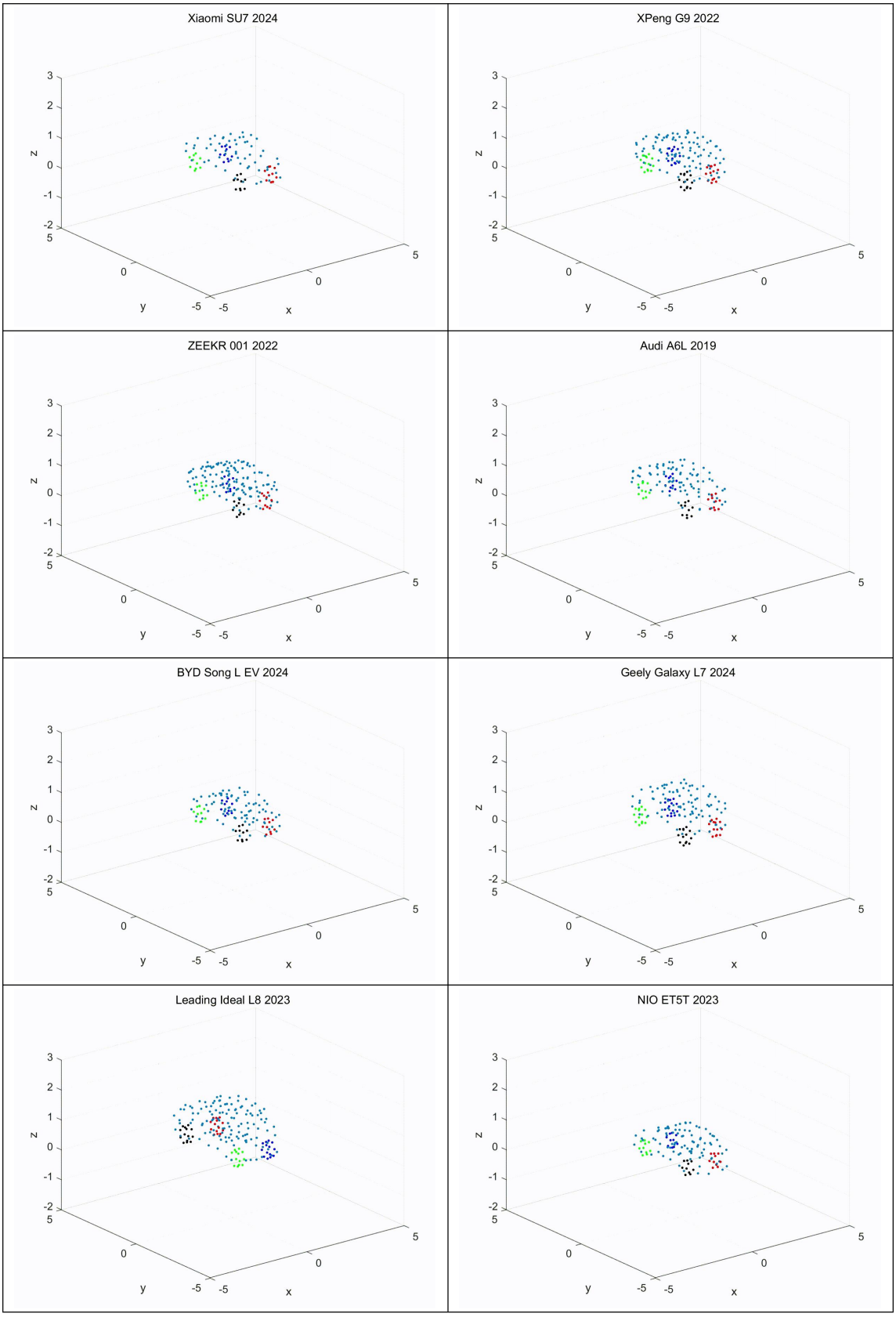
Table 3: Main information of the vehicle targets.

Name	Category	Length, Width, Height Size (m×m×m)	Wheels Diameter (m)	Forward Speed Range (m/s)
AITO M9 2024	vehicle	5.2×2.0×1.8	0.5588	1~20
Tesla Roadster 2020	vehicle	3.9×1.7×1.1	0.5080	1~20
Xiaomi SU7 2024	vehicle	5.0×2.0×1.4	0.5334	1~20
XPeng G9 2022	vehicle	4.9×1.9×1.7	0.5588	1~20
ZEEKR 001 2022	vehicle	5.0×2.0×1.6	0.5334	1~20
Audi A6L 2019	vehicle	5.0×1.9×1.5	0.5334	1~20
BYD Song L EV 2024	vehicle	4.8×2.0×1.6	0.5334	1~20
Geely Galaxy L7 2024	vehicle	4.7×1.9×1.7	0.5588	1~20
Leading Ideal L8 2023	vehicle	5.1×2.0×1.8	0.5588	1~20
NIO ET5T 2023	vehicle	4.8×2.0×1.5	0.4826	1~20

Table 3 presents the main information of vehicle targets.

The calibration point clouds of the vehicle targets are as follows.




Xiaomi SU7 2024
XPeng G9 2022
ZEEKR 001 2022
Audi A6L 2019
BYD Song L EV 2024
Geely Galaxy L7 2024
Leading Ideal L8 2023
NIO ET5T 2023
z
y
x

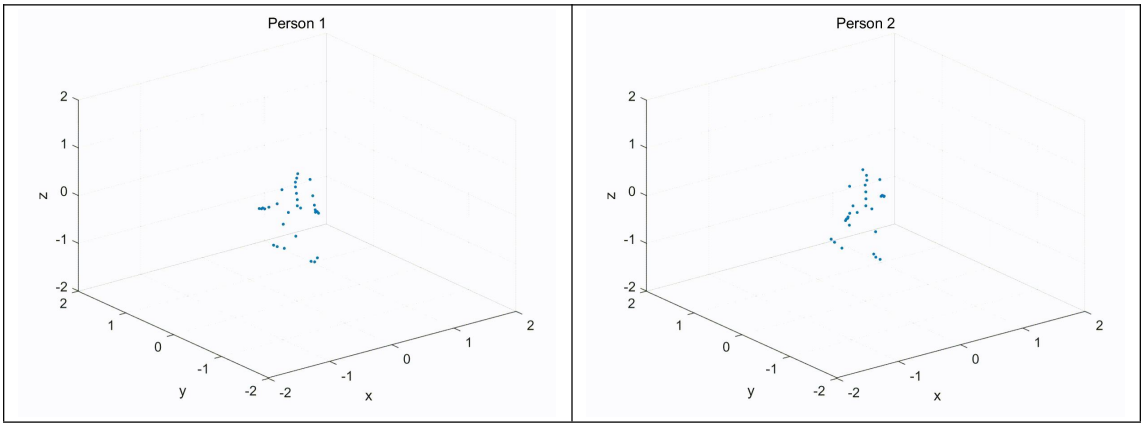
Main information of pedestrian targets.

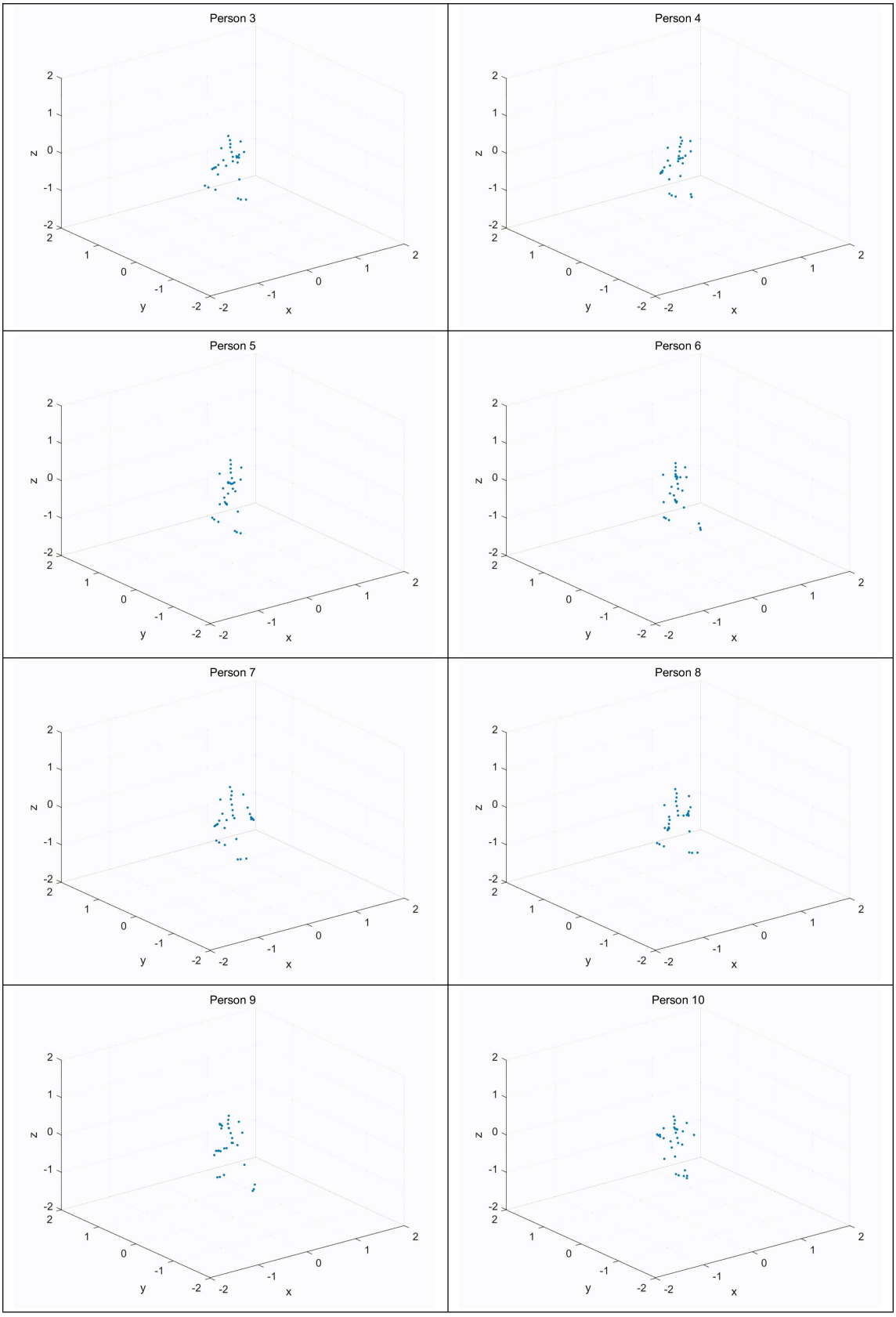
Table 4: Main information of the pedestrian targets.

Name	Category	Height (m)	Forward Speed Range (m/s)
Walking pedestrian 1	pedestrian	1.76	0.2~1.5
Walking pedestrian 2	pedestrian	1.81	0.2~1.5
Walking pedestrian 3	pedestrian	1.64	0.2~1.5
Walking pedestrian 4	pedestrian	1.67	0.2~1.5
Walking pedestrian 5	pedestrian	1.84	0.2~1.5
Walking pedestrian 6	pedestrian	1.74	0.2~1.5
Walking pedestrian 7	pedestrian	1.79	0.2~1.5
Walking pedestrian 8	pedestrian	1.66	0.2~1.5
Running pedestrian 1	pedestrian	1.76	0.2~1.5
Running pedestrian 2	pedestrian	1.65	0.2~1.5

Table 4 presents the main information of pedestrian targets.

The calibration point clouds of the pedestrian targets are as follows.




Person 3
Person 4
Person 5
Person 6
Person 7
Person 8
Person 9
Person 10
x
y
z